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BUSINESS REQUIREMENTS DOCUMENT

PDF Takeoff Automation Tool

Prepared for

Nathan Ovenden

Sydney, NSW, Australia

Document Reference

FNX-BRD-2026-NO-001

Prepared By

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Version

1.0

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Ready for Client Review

Prepared and issued by

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AI-Driven Software Development & Digital Transformation

SECTION 1

Document Control

This section records the metadata, version control, and confidentiality designation for this Business Requirements Document.

Document Title	Business Requirements Document — PDF Takeoff Automation Tool
Client	Nathan Ovenden
Prepared By	Lavesh — Fonix Technologies
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Status	Draft — Ready for Client Review
Confidentiality	Confidential — Nathan Ovenden and Fonix Technologies only

SECTION 2

Executive Summary

Nathan Ovenden operates a windows-and-doors quoting business serving the residential construction industry across New South Wales, Australia. For every new house being quoted, Nathan's team receives a submission from the customer containing architectural plans, a NatHERS energy-rating certificate, and a BASIX sustainability certificate. Buried within these documents is the list Nathan's team actually needs: every window and door in the house, with its size, type, orientation, and thermal specifications. Today, this list is extracted manually by his team, taking one to two hours per submission.

As Nathan's business grows, the manual extraction workflow is becoming the bottleneck. With over 200 submissions per month flowing through the team, the time cost is significant and the risk of human error has real financial consequences. A missed window or a wrongly transcribed dimension can translate to an inaccurate quote, a lost margin, or a customer dispute.

The PDF Takeoff Automation Tool will replace this manual workflow with a web-based application. A team member uploads the submission, the tool classifies each document type, extracts the window and door data from the appropriate source, cross-references multiple sources to flag inconsistencies, and produces a formatted Excel spreadsheet ready for quoting. What previously took one to two hours will take under five minutes. The tool is designed to handle the full range of submission formats present in Nathan's current pipeline, including digital and scanned PDFs, combined single-file submissions, and hybrid documents that contain multiple certificate types on a single page.

High-level scope summary

- Secure multi-user web application with individual accounts for Nathan's estimating team
- Document intake via web upload, with automatic classification of plans, NatHERS certificates, BASIX certificates, and hybrid submissions
- AI-driven extraction engine using Claude, with floor-plan drawings as the primary source of truth and certificates as cross-checks
- Consistency report tab surfacing any discrepancies between sources
- Formatted Excel output covering external windows, external doors, bi-fold and stacker doors, and louvres
- Revision-comparison feature producing a diff Excel when a revised plan supersedes a previous submission
- Support for complex project structures: duplexes, multi-unit developments, and multi-storey dwellings
- Admin dashboard with usage analytics, searchable one-year job history, and user management

SECTION 3

Business Context & Objectives

3.1 Business Background

Nathan Ovenden runs a New South Wales-based business that prepares windows-and-doors quotes for the residential construction sector. The business supports builders, architects, and homeowners by preparing schedules and pricing based on the plans they submit.

Currently, Nathan's team of six or more estimators receives customer submissions via email, typically containing three to five PDF files per project: the architectural plan set (30 to 80 pages), the NatHERS energy-rating certificate, the BASIX sustainability certificate, and occasionally a colour schedule or specialist energy report. The team opens each file, reads through the plan drawings and certificate tables, and manually transcribes every window and door into a quote-ready Excel spreadsheet.

This manual workflow is the current bottleneck. Each submission takes between one and two hours to process, and the team is handling over 200 submissions per month. As submission volume has grown, the error rate has also grown — a missed window or a wrong dimension in the takeoff leads to incorrect quoting downstream, with direct financial impact.

Nathan has evaluated whether to grow his team or automate the workflow. Hiring more estimators scales linearly with cost and doesn't reduce the error risk; automating the extraction scales with volume and removes the most error-prone step from the workflow entirely. This project is the automation approach.

3.2 Business Objectives

The project must deliver the following specific, measurable business outcomes:

ID	Objective	Success Metric	Priority
BO-1	Reduce time from submission receipt to quote-ready Excel	Measured as time from Upload-button-click to Excel-download-available per the audit log; target median under 5 minutes for standard submissions	Must Have
BO-2	Eliminate manual transcription errors for standard submissions	Row-level accuracy at least 95% on blind UAT sample set; accurate means extracted value matches source within ±5mm for	Must Have

ID	Objective	Success Metric	Priority
		dimensions and exact match for categorical fields (type, orientation, frame)	
BO-3	Scale to current and near-future volume without workflow impact	Sustain 200+ submissions per month with room to grow to 400+	Must Have
BO-4	Surface inconsistencies between source documents automatically	Every mismatch between plans, NatHERS, and BASIX flagged in the output without human intervention	Must Have
BO-5	Enable concurrent use by the estimating team without contention	Six or more simultaneous users able to submit and download without degraded performance	Must Have
BO-6	Reduce rework on revised submissions	When a revision supersedes an earlier plan, a diff Excel showing only the changes replaces full re-extraction	Must Have
BO-7	Provide visibility into tool usage and extraction accuracy	Admin dashboard showing jobs processed, time saved, and accuracy flags over time	Should Have

3.3 Key Stakeholders

Name / Role	Organisation	Involvement	Decision Authority
Nathan Ovenden, Owner	Client business	Primary decision-maker	Approves scope, signs off on deliverables, owns commercial decisions
Estimating Team (6+ members)	Client business	Day-to-day end users	Validate user experience, provide feedback during UAT
Lavesh, CEO & Founder	Fonix Technologies	Account owner and BA	Owns requirements, coordinates delivery, primary client contact
Fonix CTO	Fonix Technologies	Technical lead	Owns architecture and technical approach

Name / Role	Organisation	Involvement	Decision Authority
Fonix Team Lead	Fonix Technologies	Delivery lead	Owns code quality, reviews every change
Fonix QA Lead	Fonix Technologies	Quality assurance	Owns testing strategy, validates against this BRD

3.4 Project Success Criteria

The project will be considered successfully delivered when all of the following are true:

ID	Criterion	Measurement
SC-1	The end-to-end workflow (upload to Excel) completes in under 5 minutes for standard submissions	Measured on at least 20 real submissions during UAT
SC-2	Every functional requirement in Section 6 is demonstrably working	Signed-off UAT checklist mapped to this BRD
SC-3	The estimating team can operate the tool independently after training	Post-training handover sign-off by Nathan
SC-4	The tool correctly handles submissions from at least 4 different architect or builder formats	Verified against samples provided before UAT
SC-5	Consistency flags fire correctly when source documents disagree	Tested with deliberately mismatched samples during UAT
SC-6	The system remains responsive at projected peak volume	Load test simulating 10 concurrent users processing a submission each

SECTION 4

Scope Definition

4.1 In-Scope

The following functional areas are fully included in this engagement. Each is elaborated in detail in Section 6 (Functional Requirements).

Module 1: Authentication and User Management

- Individual user accounts with email and password login (per SOW response 4.2)
- Password recovery via email
- Administrator ability to create, deactivate, and manage user accounts
- Secure session management

Module 2: Document Intake and Upload

- Web-based file upload interface (per SOW response 1.5 — web upload only)
- Multi-file submission support: plans, NatHERS, BASIX, and optional supplementary documents in a single submission
- File-type validation and duplicate detection
- Upload progress tracking

Module 3: Document Classification and Routing

- Automatic detection of document type for each uploaded file (plan set, NatHERS certificate, BASIX certificate, aggregate-only energy report, hybrid single-file submission)
- Page-level analysis identifying which pages are text-based versus scanned, and which pages contain schedule data
- Automatic detection of relevant pages in 30- to 80-page plan sets (per SOW response 4.5 — auto-detect)

Module 4: Extraction Engine

- Vision-based extraction from floor plan drawings (primary source — per SOW response 3.1)
- Text-based extraction from NatHERS window and glazed door schedule (cross-check source)
- Aggregate extraction from BASIX certificate (construction summary and glazing totals)
- Canonical output schema normalising across FirstRate5, BERS Pro, Hero, and multiple architect tag formats
- Handling of both text-based and scanned PDFs, including the 20-50 percent of submissions that arrive as scans (per SOW response 1.4)

Module 5: Consistency and Validation

- Three-way reconciliation between plans, NatHERS, and BASIX
- Per-row confidence scoring
- Consistency report tab as the first sheet of every Excel output (per default)
- Job rejection pathway for low-confidence extractions (per SOW response 3.3)

Module 6: Excel Output Generation

- Single-sheet output, colour-coded by opening type (per SOW response 2.2)
- Nine data fields per row: location, window/door number, height and width, window type, orientation, glazing type, U-value and SHGC, frame material, quantity (per SOW response 2.4)
- Coverage of four opening types: external windows, external doors (entry and sliding), bi-fold and stacker doors, louvres (per SOW response 2.3)
- One row per window instance, mirroring the source schedule (per SOW response 2.5)
- Source-reference column enabling traceability from every row to its source document and page (per default)

Module 7: Revision Comparison

- Detection of revision relationships between submissions for the same project
- Diff Excel output showing windows added, removed, resized, or re-specified between revisions (per SOW response 4.4)

Module 8: Job Management and Archive

- Searchable archive of all past submissions per user and per project (per SOW response 8.1)
- One-year retention of processed jobs, source documents, and generated outputs (per SOW response 6.1)
- Re-download of any historical output

Module 9: Complex Project Handling

- Duplex projects: separate schedule per dwelling (per SOW response 7.1)
- Multi-unit developments: separate schedule per unit (per SOW response 7.1)
- Multi-storey single dwellings: schedule grouped by floor within the Excel (per SOW response 7.1)

Module 10: Admin Dashboard and Analytics

- Usage metrics: jobs processed, average time saved, accuracy flag trends (per SOW response 6.2)
- User administration (create, deactivate, manage users)
- System configuration where appropriate

4.2 Out of Scope

The following items are explicitly excluded from this engagement. Any of these can be added as a separate phase after the MVP is delivered.

Excluded opening types

- Internal doors (per SOW response 2.3)
- Roof windows and skylights (per SOW response 2.3)
- Garage doors (per SOW response 2.3)

Excluded data fields

- Opening percentage (per SOW response 2.4)
- Shading device (per SOW response 2.4)
- Manufacturer or supplier code (per SOW response 2.4)

Excluded features

- Customer-facing upload portal, where external customers upload directly (per SOW response 8.1)
- In-app editing or overriding of Excel output before download (per SOW response 8.1)
- Automatic email notifications on job completion (per SOW response 8.1)
- Mobile or tablet-optimised interfaces (per SOW response 6.3 — desktop only)

Excluded intake methods

- Email forwarding to auto-create jobs (per SOW response 1.5)
- Shared drive sync with Dropbox, Google Drive, OneDrive, or similar (per SOW response 1.5)

Excluded integrations

- Integration with existing quoting software (per SOW response 5.2)
- Integration with accounting or CRM systems
- External supplier pricing APIs or price lookup
- Building-regulation database integration

Excluded file formats

- Revit (.rvt) file support
- DWG and DXF CAD file support
- Image file submissions (.jpg, .png) outside PDF wrappers

Excluded downstream capabilities

- Automatic quote generation (this would extend the tool from takeoff to quoting; identified as a strong Phase 2 candidate)

- Customer management, CRM, or contact database
- Invoicing or payment processing

Excluded data and content work

- Historical data migration from existing systems (per standard assumption — system starts fresh)
- Content creation or copywriting
- Translation or multi-language support
- Custom-branded output design beyond standard Excel formatting

Excluded edge-case behaviours

- Mixed-case complex projects — for example, a duplex where one unit is multi-storey — are handled as the closest single pattern; any manual post-output combination is handled outside the tool
- Concurrent-session control (preventing the same user from being logged in from multiple browsers or devices) — the default is 'allowed' with no additional logic
- Revision submissions arriving before their predecessor exists in the system — these are handled as new independent submissions, not diffed
- First-login onboarding tour, tutorial, or sample-data walkthrough — handover training covers tool familiarisation instead

4.3 Assumptions

This scope, timeline, and approach depend on the following assumptions being true. If any turn out to be false, impact is documented in the right-hand column.

ID	Assumption	Impact if False
A-1	Nathan can provide additional real submissions before UAT, representing the full range of customer formats in his pipeline. Volume and timing to be confirmed with client	Extraction accuracy on edge-case formats may be lower at launch; additional post-launch refinement may be needed
A-2	Plans are treated as the source of truth over NatHERS and BASIX certificates when dimensions or specifications disagree (per SOW 3.1)	If NatHERS is the true source of truth in edge cases, reconciliation logic will need rework
A-3	Aggregate-only energy reports (e.g. TPA-style) are rare in Nathan's pipeline (per SOW 1.3)	If prevalence is higher than expected, plan-only extraction must handle more jobs end-to-end, reducing cross-check confidence

ID	Assumption	Impact if False
A-4	Over 200 submissions per month is the sustained target volume; peak bursts may exceed this but do not reflect average load (per SOW 5.1)	Infrastructure tier and Claude API cost estimates will need to be revised
A-5	A team of 6 or more estimators will use the tool, with peak concurrent usage of approximately 10 users (per SOW 4.1)	Session and worker-pool capacity may need to be scaled up
A-6	Nathan will supply branding assets (logo, colour palette) before the design phase begins	Design phase will be delayed pending receipt
A-7	A hosting approach (client-provided, Fonix-provided, or recommendation accepted) will be agreed in Section 11 before Week 0 completes	Week 0 kickoff may be delayed pending this decision
A-8	Australian data residency requirements will be confirmed before production deployment	Deployment region may need to be changed; storage costs may vary
A-9	The canonical extraction schema is validated against the UAT sample set. Architect formats not represented in UAT may require small parser extensions as they are encountered. Extensions requiring under 2 days of effort are handled within the included support window; larger additions are handled via change request	Novel formats beyond the UAT sample set may not extract correctly at launch; cost/timeline impact for large additions deferred to change request

4.4 Constraints

Constraint	Detail
Document size	Typical submission ranges from 30 to 80 pages across multiple PDF files
Scanned PDFs	Between 20 and 50 percent of submissions arrive as scans without text layers (per SOW 1.4)
Tag-format variance	Plan tag formats vary significantly between architects; no single standard across submissions
Volume	Minimum target of 200 submissions per month (per SOW 5.1)
Browser support	Modern desktop browsers only — no mobile or tablet optimisation in MVP (per SOW 6.3)
Language	English only; no multi-language support
Data retention	One year retention of processed jobs and outputs (per SOW 6.1)

4.5 Dependencies

ID	Dependency	Source	Required By	Impact if Delayed
D-1	Branding assets (logo, colour palette, typography preferences)	Client	Before design phase (Week 0)	Design phase cannot start
D-2	Additional real submissions representing full customer mix (count to be agreed with client)	Client	Before Week 2 extraction build	Extraction logic will be built against narrower sample set; accuracy at launch may be lower
D-3	Claude API access (Anthropic)	Fonix-provided	Before Week 1 development	Extraction engine cannot be built
D-4	Hosting approach decision	Client / Fonix recommendation	Before Week 0 completes	Staging and production environment setup delayed
D-5	Australian data residency confirmation	Client	Before production deployment	Deployment region and storage configuration may need rework
D-6	Nathan's time for weekly calls and end-of-week demos	Client	Throughout the build	Feedback loop slows; scope clarifications delayed
D-7	Nathan's time for UAT (estimated 3 to 5 hours spread over 3-4 days)	Client	Week 3 of build	Go-live delayed pending sign-off

SECTION 5

User Roles & Personas

5.1 User Role Matrix

The system recognises two primary user roles. A third (external customer) is explicitly out of scope in the MVP.

Role	Description	Access Level	Primary Actions	Volume
Administrator	System owner and team manager (Nathan or his office manager)	Full access to all features, all user data, all jobs, system configuration	Manage users, view usage dashboard, review flagged jobs, configure system	1 to 2
Estimator	Member of Nathan's estimating team who processes submissions day-to-day	Upload submissions, view their own job history, download outputs, view consistency flags	Upload PDF submissions, review and download Excel outputs, submit revisions, handle rejections	6 or more

5.2 User Personas

Persona 1: The Estimator

Attribute	Detail
Who they are	An estimator at Nathan's business, typically with 2 to 10 years of experience in residential construction quoting
What they need	A fast, reliable way to convert an incoming PDF submission into a quote-ready Excel spreadsheet without manual transcription
What frustrates them today	Manual extraction takes 1 to 2 hours per job, errors are embarrassing and costly, and the volume keeps growing
What success looks like	Upload a PDF, get a clean Excel in minutes, trust the output enough to quote directly from it
Technical comfort	Medium — comfortable with Excel and web tools, but not a developer. Values simplicity and predictability over advanced features

Persona 2: The Administrator

Attribute	Detail
Who they are	Nathan himself or an office manager responsible for oversight of the estimating team
What they need	Visibility into how the tool is being used, which jobs are failing, and whether the team is genuinely getting faster
What frustrates them today	No visibility into where errors happen; no way to measure whether changes to the workflow are actually helping
What success looks like	A dashboard that shows tool usage, accuracy trends, and time saved — evidence that the investment is paying off
Technical comfort	Medium — comfortable with admin interfaces but not deeply technical

5.3 Permission Matrix

Feature / Action	Administrator	Estimator	Notes
Log in with credentials	Yes	Yes	
Upload a new submission	Yes	Yes	
View own submissions history	Yes	Yes	
View all submissions across team	Yes	No	
Download Excel output	Yes	Yes	For own jobs or (admin) any job
View consistency flags and confidence	Yes	Yes	
Re-process a rejected job manually	Yes	Yes	
Delete a submission	Yes	No	Soft delete within retention window
Access admin dashboard / analytics	Yes	No	
Create or deactivate users	Yes	No	
Configure system settings	Yes	No	

Feature / Action	Administrator	Estimator	Notes
Export usage reports	Yes	No	

SECTION 6

Functional Requirements

This section is the authoritative specification of what the system must do. Each module is broken into sub-modules, each sub-module contains a table of specific requirements, each requirement is assigned a priority, a user role, and an acceptance criterion. Every decision is traceable back to Nathan's SOW question responses.

> **How to read this section**
>
> Every requirement has an ID of the form **M[Module].[Sub].[Req]**. Priority levels follow MoSCoW: **Must Have** (launch-blocker), **Should Have** (important but not blocking), and **Nice to Have** (valuable but can defer).

MODULE 1

Authentication & User Management

Module Purpose

This module secures access to the tool and ensures that each team member has an individual account. Individual accounts were explicitly chosen in SOW response 4.2 in preference to a single shared login, to enable per-user job history and usage tracking.

User roles involved: Administrator, Estimator

1.1 User Login

Authenticated access to the tool via email and password.

Req ID	Requirement	Description	Role	Priority	Acceptance
M1.1.001	Email and password login	System shall accept a registered email and password to authenticate a user	All	Must	Valid credentials grant access; invalid credentials show error
M1.1.002	Case-insensitive email	Email field shall accept any case; matching normalised to lowercase	All	Must	john@x.com and John@X.co

Req ID	Requirement	Description	Role	Priorit y	Acceptanc e
					m treated as same user
M1.1.0 03	Password visibility toggle	User can choose to show or hide their password during entry	All	Should	Eye icon toggles input type
M1.1.0 04	Remember me option	User can request their session to persist across browser restarts	All	Should	Checkbox on login extends session to 30 days
M1.1.0 05	Logout	User can end their session from any authenticated page	All	Must	Logout action invalidates session and redirects to login
M1.1.0 06	Session timeout	Inactive sessions expire after 60 minutes	All	Must	After 60 min idle, next action prompts re-login

User flow

- Step 1: User navigates to the application URL
- Step 2: System displays login page
- Step 3: User enters email and password, clicks Sign In
- Step 4: System validates credentials against user database
- Step 5: On success, user is redirected to their dashboard. On failure, inline error displayed
- Step 6: After 3 failed attempts, a 5-minute cooldown is applied to that account

1.2 Password Recovery

Self-service password reset via email.

Req ID	Requirement	Description	Role	Priority	Acceptance
M1.2.001	Forgot password link	Login page includes a Forgot Password link	All	Must	Clicking link opens reset-request flow
M1.2.002	Email reset token	On request, system sends a time-limited reset link to the registered email	All	Must	Email received within 60 seconds; link expires in 30 minutes
M1.2.003	Set new password	Using a valid reset link, user can set a new password	All	Must	New password accepted; old password no longer works
M1.2.004	Password strength	Minimum 10 characters; must include uppercase, lowercase, number, and symbol	All	Must	Weak passwords rejected with specific guidance

1.3 User Management

Administrator-only functions for managing team access.

Req ID	Requirement	Description	Role	Priority	Acceptance
M1.3.001	Invite user	Administrator can invite a new user by email. System sends an email with a set-password link	Administrator	Must	Invited user receives email within 60 seconds and can set their password

Req ID	Requirement	Description	Role	Priority	Acceptance
M1.3.002	Deactivate user	Administrator can deactivate a user account without deleting their history	Administrator	Must	Deactivated user cannot log in; their job history remains accessible to admins
M1.3.003	Reactivate user	Administrator can reactivate a previously deactivated user	Administrator	Should	Reactivation restores login ability
M1.3.004	View user list	Administrator sees a list of all users (active and deactivated) with role and last-login date	Administrator	Must	List loads within 2 seconds, searchable by name or email
M1.3.005	Change user role	Administrator can change a user's role between Estimator and Administrator	Administrator	Should	Role change takes effect on the user's next login
M1.3.006	Last-admin protection	The last active Administrator cannot be deactivated or demoted. The system prevents the action and directs the user to assign another Administrator first	System	Must	Attempting to deactivate or demote the sole active admin shows an error; action succeeds only if ≥1 active admin remains

SOW traceability: *SOW response 4.2 (individual accounts per team member) drives this module; SOW response 4.1 (6+ team members) defines expected scale.*

MODULE 2

Document Intake & Upload

Module Purpose

This module accepts PDF submissions from Nathan's team. Per SOW response 1.5, intake is web upload only — no email forwarding or shared-drive sync.

User roles involved: Administrator, Estimator

2.1 Upload Interface

Req ID	Requirement	Description	Role	Priority	Acceptance
M2.1.001	Drag and drop upload	User can drag PDF files onto a designated area to start upload	All	Must	Dropping files triggers upload immediately
M2.1.002	Click to browse	User can click to open a file browser and select files	All	Must	Standard file picker opens
M2.1.003	Multi-file upload	User can upload multiple files as a single submission (e.g., plans + NatHERS + BASIX together)	All	Must	All files uploaded are linked to one submission
M2.1.004	Upload progress	User sees progress per file with percentage and estimated time remaining	All	Must	Progress updates in real-time
M2.1.005	Cancel upload	User can cancel an upload in progress	All	Should	Cancelled uploads discard the file and free resources
M2.1.006	File-type validation	Only PDF files accepted. Disguised non-PDFs (files with .pdf extension but different format) detected via magic-byte check and rejected	All	Must	Non-PDF file rejected with clear error message

Req ID	Requirement	Description	Role	Priority	Acceptance
M2.1.007	File-size limit	Each individual PDF file up to 100 MB; total submission up to 200 MB (limits to be confirmed with client)	All	Must	Exceeding limit shows error before upload starts
M2.1.008	Password-protected PDF handling	Password-protected PDFs are rejected with a clear error message directing the user to remove the password and re-upload	All	Must	Password-protected file shows: 'This file is password-protected. Please remove the password and re-upload'
M2.1.009	Upload interruption recovery	If an upload is interrupted (network failure, browser closure), the partial upload is discarded cleanly. User can restart the upload on return without orphaned records	All	Must	Interrupted upload does not create job records or consume quota; clean restart available

2.2 Submission Metadata

Req ID	Requirement	Description	Role	Priority	Acceptance
M2.2.001	Project name	User can assign a project name or reference to the submission	All	Must	Project name saved with submission
M2.2.002	Customer reference	User can assign an optional customer reference to the submission	All	Should	Reference saved and searchable later

Req ID	Requirement	Description	Role	Priorit y	Acceptanc e
M2.2. 003	Auto-generat ed ID	System generates a unique ID per submission if no project name is entered	All	Must	Every submission has a unique ID format: FNX-YYYY MMDD-N NN
M2.2. 004	Notes field	Optional free-text notes on the submission	All	Nice to have	Notes saved and displayed in job history

2.3 Processing Experience

After upload, extraction runs in the background. This sub-module defines what the user experiences during that wait.

Req ID	Requirement	Description	Role	Priorit y	Acceptanc e
M2.3. 001	Processing view	After upload completes, the user sees a processing-in-progress view showing job ID, queue position (if any), and estimated time remaining	All	Must	Processing view appears within 2 seconds of upload completion
M2.3. 002	Browser-clos e resilience	The user may navigate away or close the browser during processing. The job continues on the server and appears in the user's job history when complete	All	Must	Closing the browser does not halt or lose the job
M2.3. 003	Concurrent submissions	The user can submit additional jobs while earlier jobs are still processing, up to a per-user concurrent-job limit (default 5, tuneable by admin)	All	Should	Up to 5 concurrent jobs allowed; 6th queued or shown 'at limit' message

Req ID	Requirement	Description	Role	Priority	Acceptance
M2.3.004	Completion indicator	When processing completes, the user sees a visual indicator on their dashboard on their next view. No email notification is sent (per SOW 8.1 exclusion)	All	Must	Completed job appears highlighted on dashboard until dismissed or viewed

SOW traceability: *SOW response 1.5 confirms web upload only; email forwarding, shared drive sync, and customer-facing portals are out of scope for MVP. SOW 8.1 excludes email notifications on completion, replaced here with in-app indicator.*

MODULE 3

Document Classification & Routing

Module Purpose

When a user uploads one or more PDFs, the system must automatically determine what each document is (plan, NatHERS, BASIX, combined) so it can route extraction to the right engine. This module handles that classification.

User roles involved: System (automated); Administrator can override classifications

3.1 Document-Type Detection

Req ID	Requirement	Description	Role	Priority	Acceptance
M3.1.001	Classify architectural plans	System identifies PDFs that are architectural plan sets	System	Must	Correct classification on at least 18 of 20 test submissions (≥90%)
M3.1.002	Classify NatHERS certificate	System identifies PDFs that are NatHERS certificates across all three software producers (Hero, BERS Pro, FirstRate5)	System	Must	All three formats recognised correctly

Req ID	Requirement	Description	Role	Priority	Acceptance
M3.1.003	Classify BASIX certificate	System identifies NSW BASIX certificates	System	Must	BASIX format recognised correctly
M3.1.004	Identify hybrid single-file	System detects when a single PDF contains multiple document types (e.g., plan sheet with BASIX stamp and NatHERS pages appended)	System	Must	Hybrid files split for section-by-section processing
M3.1.005	Detect aggregate-only reports	System identifies TPA-style aggregate-only energy reports and routes them to plan-based extraction (per SOW 3.2)	System	Must	Aggregate-only reports correctly flagged; plan extraction engaged
M3.1.006	Unknown document handling	System flags any uploaded document that doesn't match known types	System	Must	Unknown type shown in consistency report and job summary
M3.1.007	Plans-only submission	When a submission contains plans but neither NatHERS nor BASIX, the system attempts plan-based extraction and produces a consistency report with explicit warning that no cross-check sources were available	System	Must	Plans-only submission produces an Excel clearly flagged 'no cross-check sources'

3.2 Page-Level Analysis

Req ID	Requirement	Description	Role	Priority	Acceptance
M3.2.001	Detect scanned pages	For each page, system determines whether the page has a usable text layer or is a scanned image	System	Must	Scan-detection accurate on mixed test set
M3.2.002	Identify schedule pages	In plan sets, system identifies which pages contain window or door schedules (tables or callout-dense floor plans)	System	Must	Relevant pages surfaced with confidence score
M3.2.003	Identify elevation vs floor plan pages	System distinguishes elevation drawings from floor plans	System	Should	Floor plans preferred for tag extraction
M3.2.004	Auto-page-selection	System processes relevant pages automatically without user page selection (per SOW 4.5)	System	Must	User uploads full PDF; no manual page picking required

SOW traceability: *SOW responses 1.2, 1.3, 1.4, 3.2, 4.5 shape this module. Nathan's submission mix is plans+NatHERS+BASIX, aggregate-only reports are rare, 20-50% are scanned, and auto-detection of pages is preferred over manual selection.*

MODULE 4

Extraction Engine

Module Purpose

The core intelligence of the system. This module reads each document, pulls out window and door data, and normalises it into a canonical internal format. Critically, per SOW response 3.1, plans are treated as the primary source of truth when they disagree with the certificates — not the other way around.

User roles involved: System (automated)

Scope of extraction coverage

The extraction engine is built and tested against the architect and assessor formats represented in the UAT sample set. Formats not represented are handled on a best-effort basis; new format extensions are added via the change request process described in Section 14. This sets a clear boundary between 'in scope' formats (the sample set) and 'format extensions' (anything new that appears later).

4.1 Plan-Based Extraction (Primary)

Req ID	Requirement	Description	Role	Priority	Acceptance
M4.1.001	Vision-based extraction	System uses Claude vision API to extract window and door callouts from floor plan pages	System	Must	Row-level accuracy ≥95% on blind UAT sample set of ≥15 submissions (dimensions within ±5mm of source, exact match for categorical fields)
M4.1.002	Handle rotated text	System correctly interprets tags written in any orientation on the plan	System	Must	Rotated labels extracted correctly
M4.1.003	Tag-format normalisation	System handles multiple architect tag styles (e.g., W3 1800H x 610W, DSD2115, 2136ALSD, 18-09AAW-DG-LOWE) and normalises to the canonical schema	System	Must	All four sample architect formats extract correctly
M4.1.004	Scanned-page vision	When a page is detected as scanned, system rasterises at 150 DPI and sends directly to vision extraction	System	Must	Scanned plans produce comparabl

Req ID	Requirement	Description	Role	Priority	Acceptance
					e extraction to digital
M4.1.005	Extract location/room	Each window or door is linked to the room it belongs to based on proximity on the floor plan	System	Must	Per-row basis: ≥90% of extracted rows show correct room reference vs manual QA interpretation on UAT sample set
M4.1.006	API error handling	If the Claude API returns an error, times out, or produces malformed data, the system retries up to 3 times with exponential backoff. After 3 failures, the job is marked Failed (distinct from Rejected) and the user is notified with an option to retry	System	Must	On injected API failure, the job reaches Failed status with user-visible retry action

4.2 NatHERS-Based Extraction (Cross-Check)

Req ID	Requirement	Description	Role	Priority	Acceptance
M4.2.001	Parse window schedule	System extracts the itemised window and glazed door schedule from the NatHERS certificate	System	Must	Every row of the schedule extracted with all fields
M4.2.002	Parse performance table	System extracts U-value and SHGC per window type	System	Must	Performance data linked to each

Req ID	Requirement	Description	Role	Priorit y	Acceptanc e
					window instance
M4.2. 003	Handle three software formats	Extraction works across Hero, BERS Pro, and FirstRate5 NatHERS outputs	System	Must	All three formats parse correctly
M4.2. 004	Use as cross-check	Per SOW 3.1, NatHERS data is used as a cross-check against plan data, not as the primary source	System	Must	Reconciliat ion logic treats plan as authoritati ve

4.3 BASIX-Based Extraction (Aggregate Check)

Req ID	Requirement	Description	Role	Priorit y	Acceptanc e
M4.3. 001	Parse construction table	System extracts the construction table (wall, roof, floor, insulation) from BASIX	System	Must	Full constructio n table captured
M4.3. 002	Parse glazing aggregates	System extracts frame-material and glazing-type totals	System	Must	Totals captured correctly for reconciliati on
M4.3. 003	Use as aggregate check	BASIX aggregates reconciled against sum of plan-derived windows	System	Must	Mismatche s flagged in consistenc y report

4.4 Canonical Schema

Req ID	Requirement	Description	Role	Priorit y	Acceptanc e
M4.4. 001	Unified output schema	All extraction sources produce data in a single canonical schema with 9 fields: location,	System	Must	All sources produce uniformly

Req ID	Requirement	Description	Role	Priorit y	Acceptanc e
		W/D number, H x W, type, orientation, glazing, U-value/SHGC, frame material, quantity (per SOW 2.4)			shaped data
M4.4. 002	Opening-typ e filtering	Only the 4 opening types Nathan requires are extracted: external windows, external doors, bi-fold/stacker, louvres (per SOW 2.3)	System	Must	Internal doors, roof windows, garage doors ignored
M4.4. 003	One-row-per -instance	When the same window appears multiple times, each instance is a separate row (per SOW 2.5)	System	Must	W10 and W11 with identical specs shown as 2 rows
M4.4. 004	NatHERS terminology	Window types use NatHERS standard terms (Awning, Sliding Door, Fixed, Double Hung, etc.) per SOW 2.6	System	Must	Terms consistent across all outputs

SOW traceability: *SOW response 3.1 defines plans as primary. SOW 2.3, 2.4, 2.5, 2.6 shape the output schema. SOW 1.4 requires handling scanned documents. SOW 3.2 requires plan-only extraction when NatHERS is aggregate-only.*

MODULE 5

Consistency & Validation

Module Purpose

Once data has been extracted from multiple sources, this module cross-checks them for agreement and surfaces any disagreements for human review. Per SOW response 3.3, if confidence is too low or sources are severely contradictory, the job is rejected rather than silently producing a bad Excel.

User roles involved: System (automated); Estimator and Administrator consume the reports

How the automatic and rejection paths work together

SOW response 3.3 asked for rejection of low-confidence jobs, and SOW response 4.3 asked for the workflow to be fully automatic. The BRD reconciles these as a bifurcation:

- Submissions above the rejection threshold are processed fully automatically and produce a quote-ready Excel. The estimator's team edits in Excel afterwards if needed (SOW 4.3).
- Submissions below the rejection threshold are routed to the Rejected Queue for manual handling, rather than silently producing a bad Excel (SOW 3.3).

This satisfies both answers simultaneously.

5.1 Three-Way Reconciliation

Req ID	Requirement	Description	Role	Priority	Acceptance
M5.1.001	Plans as primary source	When plans and NatHERS disagree on dimension, type, or location, plan data is used and disagreement is flagged (per SOW 3.1)	System	Must	Reconciliation logic prefers plan data in conflicts
M5.1.002	Cross-check with NatHERS	NatHERS data compared to plan data per window; mismatches flagged	System	Must	Every NatHERS-plan mismatch captured
M5.1.003	Aggregate check against BASIX	Sum of plan-derived windows reconciled against BASIX aggregate totals; mismatches flagged	System	Must	BASIX-plan delta reported
M5.1.004	Missing-in-NatHERS flag	If a window appears in plans but not NatHERS, or vice versa, it's flagged	System	Must	Missing items surfaced in consistency report

5.2 Confidence Scoring

Req ID	Requirement	Description	Role	Priority	Acceptance
M5.2.001	Per-row confidence	Every extracted row has a confidence score (0-100) based on source data quality, cross-check agreement, and extraction certainty	System	Must	Confidence visible on each row
M5.2.002	Low-confidence threshold	Rows with confidence below 70% are highlighted in the Excel output	System	Must	Low-confidence rows visually distinct
M5.2.003	Submission-level confidence	Each submission gets an overall confidence score (average of rows, weighted by severity of flags)	System	Must	Overall score displayed in job summary

5.3 Rejection Workflow

Req ID	Requirement	Description	Role	Priority	Acceptance
M5.3.001	Auto-reject low confidence jobs	If overall submission confidence is below 50%, or more than 25% of rows are low-confidence (applied only when ≥5 rows are evaluated; trivial submissions with <5 rows use overall confidence alone), the job is rejected rather than producing an Excel (per SOW 3.3). Thresholds are initial defaults and tuneable during UAT	System	Must	Low-confidence jobs clearly flagged as Rejected; trivial submissions do not auto-reject on percentage rule alone
M5.3.002	Notification of rejection	User is notified in the UI when a job is rejected, with reason	System	Must	Rejection reason visible in job summary
M5.3.003	Manual re-processing option	Rejected job can be manually marked for review by Estimator or Administrator	Estimator,	Must	Rejected jobs can be reopened

Req ID	Requirement	Description	Role	Priority	Acceptance
			Administrator		

5.4 Consistency Report

Req ID	Requirement	Description	Role	Priority	Acceptance
M5.4.001	First tab of Excel	Consistency report appears as the first tab of every Excel output (per default)	System	Must	Consistency tab always first
M5.4.002	List all flags	Report itemises every flag: source disagreements, missing items, low confidence rows	System	Must	Every flag listed with window reference and description
M5.4.003	Summary counters	Report shows counts: total windows, total flags, confidence distribution	System	Must	Summary section at top of report
M5.4.004	Actionable descriptions	Each flag includes a clear description of what's wrong and where to verify	System	Must	Flag text is a single sentence that (a) identifies the window reference, (b) describes the discrepancy, (c) specifies the source document(s) to verify against

Req ID	Requirement	Description	Role	Priority	Acceptance
M5.4.005	Enumerated flag categories	The consistency report surfaces exactly the following categories: Missing-in-plans, Missing-in-NatHERS, Dimension mismatch, Orientation mismatch, Window-type mismatch, Frame-material mismatch, BASIX-aggregate mismatch, Low-confidence row, and Unreadable source page	System	Must	Every flag raised in UAT falls into one of the nine categories; no uncategorised flags

5.5 Rejected-Job Review Queue

When the system rejects a job, the submission does not disappear. It is routed to a Rejected Queue where Administrators and Estimators can resolve it.

Req ID	Requirement	Description	Role	Priority	Acceptance
M5.5.001	Rejected Queue view	Rejected jobs appear in a dedicated Rejected Queue accessible to Administrators (all rejects across team) and to the submitting Estimator (own rejects only)	Administrator, Estimator	Must	Queue lists all rejected jobs with timestamp, reason, and submitter
M5.5.002	Queue actions	From the Rejected Queue, a user can: (a) manually mark the job for re-processing, (b) mark the job as Resolved (indicating it was handled via manual takeoff outside the system), or (c) archive the job	Administrator, Estimator	Must	Each action available with one click; status updates accordingly
M5.5.003	Resolution audit	Every action on a rejected job is recorded in the audit log with user, timestamp, and action taken	System	Must	Audit log entry created for every Rejected-Queue action

SOW traceability: *SOW response 3.1 sets reconciliation direction. SOW 3.3 mandates rejection of low-confidence jobs rather than silent bad output. Consistency report is a committed default. The Rejected Queue closes the loop on SOW 3.3 so rejected jobs have an explicit handling path.*

MODULE 6

Excel Output Generation

Module Purpose

This module produces the final deliverable for every successful submission: a formatted Excel file that's immediately usable in Nathan's quoting workflow.

User roles involved: System generates; Estimator downloads

6.1 Excel Structure

Req ID	Requirement	Description	Role	Priority	Acceptance
M6.1.001	Single-sheet takeoff per dwelling	For a standard single-dwelling project, takeoff data appears on a single sheet, colour-coded by the four included opening types: external windows, external doors, bi-fold and stacker doors, and louvres (per SOW 2.2 and 2.3). For multi-dwelling projects, see M6.1.007	System	Must	Single-dwelling Excel has one takeoff sheet with 4 colour-coded opening types visually distinguishable
M6.1.002	Nine data columns	Columns: Location, W/D Number, Height (mm), Width (mm), Type, Orientation, Glazing Type, U-value / SHGC, Frame Material, Quantity (per SOW 2.4)	System	Must	All 9 columns present with correct data
M6.1.003	Frozen top row	Header row frozen for scrolling	System	Must	Headers remain visible on scroll
M6.1.004	Colour-coded headers	Each opening type (window, door, bi-fold, louvre) has a distinct header colour	System	Must	Colours distinct and accessible

Req ID	Requirement	Description	Role	Priority	Acceptance
M6.1.005	Dimensions in mm	All dimensions expressed in millimetres (per default, matching NatHERS and Australian plans)	System	Must	No conversion issues
M6.1.006	Low-confidence highlighting	Rows below confidence threshold highlighted (per M5.2.002)	System	Must	Highlight colour distinct from data colours
M6.1.007	Multi-dwelling output structure	For multi-dwelling projects (duplexes and multi-unit), the output is a single Excel workbook containing the consistency report at tab position 1, followed by one takeoff sheet per dwelling or unit. Sheet names: Dwelling 1, Dwelling 2 (duplex) or Unit 1, Unit 2 … Unit N (multi-unit). For multi-storey single dwellings, grouping by floor happens within one sheet (see Module 9)	System	Must	Duplex output contains 1 consistency + 2 dwelling sheets; multi-unit contains 1 consistency + N unit sheets

6.2 Source Traceability

Req ID	Requirement	Description	Role	Priority	Acceptance
M6.2.001	Source-reference column	Every row includes a Source column showing which document and page the data came from (per default)	System	Must	Reference like Plan p.3 or NatHERS p.6 on each row
M6.2.002	Tab: consistency report	First tab of the workbook is the consistency report (per default)	System	Must	Report always first tab

6.3 Download and Delivery

Req ID	Requirement	Description	Role	Priority	Acceptance
M6.3.001	In-app download	User downloads the Excel directly from the job summary page (per default)	All	Must	Download completes in under 10 seconds
M6.3.002	Re-download historical	User can re-download Excel from any past job in their history	All	Must	Past outputs available for 1 year
M6.3.003	File naming convention	Excel files named using a consistent pattern (e.g., ProjectName_FNX-YYYYMMDD-NNN.xlsx)	System	Should	Naming consistent and sortable

SOW traceability: *SOW responses 2.2, 2.3, 2.4, 2.5, 2.6 define output structure. Source column, consistency report, dimensions in mm are all committed defaults per the defaults appendix of the SOW.*

MODULE 7

Revision Comparison (Diff Excel)

Module Purpose

When a customer sends a revised plan (e.g., REV B after REV A), this module detects the revision relationship and produces a diff Excel showing what changed. Selected as must-have in SOW response 4.4.

User roles involved: Estimator primary; System automates detection and diff generation

7.1 Revision Detection

Per SOW response 4.4, the tool generates a diff Excel when a submission supersedes an earlier one. The mechanism by which revisions are linked to prior submissions is to be confirmed with the client during Week 0 design. The requirement below captures the minimum viable approach; the UX will be finalised in the design phase.

Req ID	Requirement	Description	Role	Priority	Acceptance
M7.1.001	Manual revision link	User can explicitly mark a new submission as a revision of an existing one in the same project	Estimator	Must	Link establishes revision

Req ID	Requirement	Description	Role	Priorit y	Acceptanc e
					relationshi p; diff Excel produced on next run

7.2 Diff Generation

Req ID	Requirement	Description	Role	Priorit y	Acceptanc e
M7.2.0 01	Diff Excel	Output includes a diff Excel showing added, removed, resized, and respecified windows/doors (per SOW 4.4)	System	Must	Every changed row is classified into exactly one of four mutually-e xclusive categories (Added, Removed, Resized, Respecifie d); category visible in a dedicated Change Category column
M7.2.0 02	Visual diff markers	Added rows in green, removed in red, changed in amber	System	Must	Colour coding consistent
M7.2.0 03	Full new Excel	Alongside the diff, the full new Excel is also available	System	Must	Both files download able from job summary

SOW traceability: *SOW response 4.4 places revision-comparison in MVP rather than Phase 2. This adds approximately one week to the build timeline and is reflected in the Scope of Work commercial terms. Scope assumption: window numbering is stable across revisions. Scenarios* *where the architect fully renumbers windows between revisions are out of scope for MVP and are handled as new submissions rather than diffed.*

MODULE 8

Job Management & Archive

Module Purpose

This module gives Estimators access to the history of their work and provides administrators with oversight of the team's activity.

User roles involved: Estimator (own jobs), Administrator (all jobs)

8.1 Job Listing and Search

Req ID	Requirement	Description	Role	Priorit y	Acceptanc e
M8.1.0 01	My jobs list	Estimator sees a list of their own jobs with date, project name, status, confidence	Estimat or	Must	List loads within 2 seconds
M8.1.0 02	All jobs list (admin)	Administrator sees all jobs across the team	Adminis trator	Must	Team-wid e view
M8.1.0 03	Search, filter, and sort	The archive is searchable across project name, customer reference, and date range, with filters by status and sortable by date or project (per SOW 8.1)	All	Must	Any past submission locatable in three interaction s or fewer; search returns results within 1 second

8.2 Job Detail View

Req ID	Requirement	Description	Role	Priorit y	Acceptanc e
M8.2. 001	Summary panel	View shows submission metadata, extraction summary, confidence, flag count	All	Must	All key info visible
M8.2. 002	Download Excel	Excel file downloadable from detail view	All	Must	Download works

Req ID	Requirement	Description	Role	Priority	Acceptance
M8.2.003	Download source docs	Original source PDFs downloadable	All	Should	Source files preserved
M8.2.004	View consistency flags	Flag list visible in the UI (same data as Excel consistency tab)	All	Must	Flags displayed

8.3 Retention

Req ID	Requirement	Description	Role	Priority	Acceptance
M8.3.001	1-year retention	Processed jobs, source files, and outputs retained for 1 year (per SOW 6.1)	System	Must	Data retained 1 year; behaviour after 1 year to be confirmed with client
M8.3.002	Manual delete	Administrator can manually delete a job at any time	Administrator	Should	Delete action available from job detail view

8.4 Submission Deletion

Supports the Delete permission in the Section 5.3 permission matrix.

Req ID	Requirement	Description	Role	Priority	Acceptance
M8.4.001	Admin soft delete	Administrators can delete any submission. Deletion is a soft-delete: the record is marked deleted and removed from default views, retained until the retention period ends, after which it is permanently purged	Administrator	Must	Deleted submission no longer appears in default lists; purge occurs at retention end

Req ID	Requirement	Description	Role	Priority	Acceptance
M8.4.002	No estimator delete	Estimators cannot delete submissions (per permission matrix)	System	Must	Delete action hidden or disabled for Estimator role
M8.4.003	Source-file lockout on delete	On deletion, source files and outputs become inaccessible immediately, even though the record is retained until purge	System	Must	Download attempts on deleted submission return 'not available' message

8.5 Submission Status Lifecycle

Every submission progresses through a defined lifecycle. The status at any point determines what actions are available to the user.

Status	Meaning	Valid next states
Uploaded	Files received; job queued for processing	Processing
Processing	Extraction running	Completed, Failed, Rejected
Completed	Extraction succeeded; Excel available for download	(terminal; can be Deleted)
Rejected	Auto-rejected due to low confidence or contradictory sources (per Module 5.3)	Processing (on manual re-processing), Resolved (on manual close)
Failed	Technical failure (API error, corrupted PDF, etc.)	Processing (on retry), (terminal otherwise)
Resolved	Rejected job closed manually after handling outside the system	(terminal; can be Deleted)
Deleted	Admin soft-delete; awaiting purge at retention end	(terminal; purged at retention)

SOW traceability: *SOW response 6.1 defines 1-year retention. SOW 8.1 includes archive/history as MVP must-have.*

MODULE 9

Complex Project Handling

Module Purpose

Not every submission is a single simple dwelling. Per SOW response 7.1, Nathan's pipeline regularly includes duplexes, multi-unit developments, and multi-storey single dwellings. Each requires specific output formatting. At upload time, the estimator indicates which complex-project pattern (if any) applies; the system then formats the output accordingly.

User roles involved: Estimator (selects project type at upload); System (formats output)

9.1 Output Format per Project Type

Req ID	Requirement	Description	Role	Priority	Acceptance
M9.1.001	Duplex output	For duplexes (two dwellings on one site), system produces a separate schedule per dwelling (per SOW 7.1)	System	Must	Two schedules clearly labelled Dwelling 1 and Dwelling 2
M9.1.002	Multi-unit output	For multi-unit developments (3 or more units), system produces a separate schedule per unit (per SOW 7.1)	System	Must	One schedule per unit clearly labelled
M9.1.003	Multi-storey grouping	For multi-storey single dwellings, the schedule is grouped by floor within the Excel (per SOW 7.1)	System	Must	Floor grouping visible; header rows identify each floor

SOW traceability: *SOW response 7.1 ticks exactly three patterns (duplex separate, multi-unit separate, multi-storey grouped). This module maps one-to-one to those three ticks. Project-type selection at upload (vs automatic detection) is the simplest and most reliable approach and will be confirmed during Week 0 design. The specific output structure (separate sheets vs separate files vs single sheet with Unit/Floor column) will also be confirmed.*

MODULE 10

Admin Dashboard & Analytics

Module Purpose

Administrative oversight of the tool's usage, performance, and accuracy over time.

User roles involved: Administrator

10.1 Usage Metrics

Req ID	Requirement	Description	Role	Priority	Acceptance
M10.1.001	Jobs processed	Dashboard shows total jobs processed (daily, weekly, monthly)	Administrator	Should	Metric displayed and trending
M10.1.002	Time saved	Estimate of time saved (jobs × avg manual time) displayed over time	Administrator	Should	Metric displayed
M10.1.003	Accuracy flags over time	Rejection rate, confidence distribution, flag frequency trending (per SOW 6.2)	Administrator	Should	Trend chart visible
M10.1.004	Per-user activity	Usage broken down by Estimator	Administrator	Should	Per-user metrics visible

SOW traceability: *SOW response 6.2 selects admin dashboard in MVP, confirmed with client. Scope is deliberately a basic dashboard covering the four metrics above; advanced analytics, CSV export, and per-metric drill-downs are Phase 2.*

SECTION 7

Non-Functional Requirements

7.1 Performance

Definition of standard vs edge submission

- Standard submission: plan set ≤40 pages, digital PDF with a text layer, single dwelling

- Edge submission: 41+ pages, or scanned PDF (no text layer), or multi-dwelling, or ≥20% of extracted rows below confidence threshold

Performance targets apply to standard submissions. Edge submissions are handled on a best-effort basis within the technical buffer and are not subject to the same SLA.

Requirement	Target	Measurement
Page load time	Under 3 seconds on standard broadband	Lighthouse / Chrome DevTools
End-to-end job processing (standard)	Target under 5 minutes (not a hard SLA; subject to confirmation during UAT)	Measured in production against agreed target
End-to-end job processing (edge)	Best-effort; no committed target	Observed range reported during UAT
Concurrent users	10+ simultaneous users without degradation	Load tested pre-launch
API response time (non-extraction)	Under 500 ms for standard requests	Server-side monitoring
Job queue capacity	200+ submissions per month sustained (per SOW 5.1)	Queue depth monitoring

7.2 Security

Requirement	Description
Authentication	Email and password with bcrypt-hashed passwords; session tokens with 60-minute inactivity timeout (per SOW 4.2)
Encryption in transit	All client-server traffic over HTTPS / TLS 1.3
Encryption at rest	Uploaded PDFs and generated Excel files encrypted at rest (AES-256 or cloud-native equivalent)
Input validation	All user inputs validated and sanitised on both client and server side
Password policy	Minimum 10 characters; must include uppercase, lowercase, number, and symbol

Requirement	Description
Access control	Role-based access enforced per Section 5.3 permission matrix
Audit logging events	The following events are logged with user, timestamp, and relevant metadata: login, logout, failed login, submission upload, submission download (Excel or source PDF), job deletion, rejected-queue action (re-process, resolve, archive), user invitation, user deactivation, user role change, system configuration change
Audit log retention	Retention aligned with job-data retention (1 year) unless otherwise agreed with client
File upload security	Magic-byte validation; password-protected PDFs rejected at upload (M2.1.008); uploads isolated from the execution environment
API cost controls	Per-user daily submission cap (default 20, admin-configurable). Monthly Claude API spend alert fires at 80% of projected budget; hard-stop at 120% of projected budget pending admin review
PDF lifecycle	Uploaded PDFs are retained alongside the job record for the retention period and encrypted at rest. On deletion, PDFs are removed from active storage immediately; backup copies purged within 30 days

7.3 Browser & Device Support

Platform	Supported Versions
Chrome	Latest 2 versions
Firefox	Latest 2 versions
Safari	Latest 2 versions
Edge	Latest 2 versions
iOS Safari	Not supported in MVP (per SOW 6.3)
Android Chrome	Not supported in MVP (per SOW 6.3)
Responsive breakpoints	Desktop (1440px+) and laptop (1024px) only — no tablet or mobile layouts in MVP

7.4 Accessibility

WCAG 2.1 AA is referenced as the minimum accessibility standard for this tool, even where not a formal contractual requirement. The following practices apply:

- Keyboard navigation supported for all core workflows (upload, review, download, admin actions)
- Semantic HTML with appropriate ARIA labels on all interactive controls
- Colour contrast ratio ≥4.5:1 for body text, ≥3:1 for large text and meaningful graphics
- Visible focus indicators on all interactive elements

Full WCAG 2.1 AA certification, screen-reader optimisation, and assistive-technology testing are not in scope.

7.5 Localisation

Requirement	Details
Languages supported	English only
Date format	DD/MM/YYYY (Australian convention)
Time format	24-hour with AEST timezone display
Currency	Not applicable — tool does not handle monetary values
Timezone handling	AEST for all user-facing timestamps; UTC for internal storage

7.6 Availability, Reliability & Monitoring

Requirement	Target
Uptime	Target to be confirmed with client during Week 0; the system will be architected for standard business-hours reliability
Backup frequency	Daily automated backups; point-in-time recovery for database
Disaster recovery	Recovery Time and Recovery Point Objectives to be confirmed with client based on their tolerance for downtime and data loss
Planned maintenance windows	Outside Australian business hours; announced 48 hours in advance
Monitoring	Automated uptime and error-rate monitoring; Fonix on-call notified within 5 minutes of a service-impacting incident
Client incident communication	Nathan is notified via the agreed channel within 1 hour of Fonix confirming a service-impacting outage

7.7 Scalability

Requirement	Details
Expected users at launch	6-10 active users (per SOW 4.1)
Expected submission volume at launch	200+ per month (per SOW 5.1)
Future growth	Volume growth expected but not specifically projected; architecture supports horizontal scaling
Scaling approach	Horizontal: web app auto-scales based on request load; worker pool scales based on job queue depth

7.8 Data Export & Portability

Nathan's data belongs to Nathan. The system provides an export capability so he can retrieve a complete copy at any time, including if he chooses to migrate away from the tool in future.

Requirement	Details
Exportable data	Job metadata (CSV); source PDFs (ZIP); Excel outputs (ZIP); user records (CSV); audit log (CSV)
Trigger	Administrator initiates export from the admin settings area
Delivery	Complete export bundled as a single ZIP download; large exports processed asynchronously with email-link delivery (the only system email, to admin only)
Performance target	Export completes within 10 minutes for up to 1 year of data at projected volume
Frequency	On-demand; no scheduled exports in MVP

SECTION 8

Third-Party Integrations

This project uses one primary third-party integration. Additional integrations (CRM, quoting software, supplier APIs) are explicitly out of scope per Section 4.2.

ID	Service	Provider	Purpose	Direction	Complexity
INT-1	Claude API	Anthropic	AI-driven document extraction (text and vision)	Send (requests) / Receive (responses)	Medium — established API, well-documented

8.1 INT-1: Claude API (Anthropic)

Attribute	Detail
What it does	Processes images and text from uploaded PDFs to extract window and door data, classify documents, and assist with reconciliation logic
Data flow	Fonix system sends page images and prompts to Claude; Claude returns structured extraction results
Data transmitted	Page images (rasterised PDF pages), extracted text, and metadata. Does not include user PII beyond what's embedded in the architectural plans
Failure handling	On API failure: automatic retry with exponential backoff; after 3 failures, job marked as Failed and user notified. Queued jobs resume when API recovers
Cost structure	Pay-per-call based on input and output token count; estimated $1-3 USD per submission at current pricing (200+ submissions/month = $200-600 USD/month in API cost)
Client responsibility	None — Fonix provides and manages Claude API access, API cost will be disclosed transparently as part of monthly running cost

SECTION 9

Data Requirements

9.1 Data Sources

Source	Type	Format	Volume	Migration
Uploaded PDF submissions	Binary files	PDF (text or scanned)	200+/month, typically 1-10 MB each	No migration — new submissions only
User accounts	Structured	Database records	6-15 active users	No migration — created fresh
Generated Excel outputs	Binary files	XLSX	200+/month matching submission count	No migration
Audit logs	Structured	Database records	All user actions, retained 1 year	No migration

9.2 Data Migration

No data migration is required. Per standard assumption, the system launches empty and new submissions begin on Day 1 of production. Any historical extractions already produced manually by Nathan's team remain in his existing files and are not imported into the new tool.

9.3 Content Requirements

Content Type	Provided By	Format	Deadline
Logo and brand assets	Client	SVG or high-resolution PNG with transparency	Before design phase starts
Colour palette	Client	Hex codes or brand guidelines document	Before design phase starts
Sample submissions	Client	Real PDFs representing the range of customers and scenarios (count to be agreed)	Before Week 2 extraction build
Test data for UAT	Client	3-5 additional real submissions not previously shown	Before UAT

SECTION 10

Design Requirements

10.1 Design Approach

Aspect	Detail
Methodology	Desktop-first (per SOW 6.3 — mobile not required in MVP)
Design tool	Figma — deliverables include full interactive prototype
Deliverables	Wireframes, high-fidelity screens for all user-facing flows, component library, interaction states (default, hover, active, loading, empty, error)
Revision rounds	Two rounds included. A round is defined as: one formal review of the agreed design set by the client, followed by consolidated written feedback within 5 business days. Additional rounds via change request
Design timing	Design phase runs in Week 0 / Week 1, approved before development of any given module

10.2 Branding

Aspect	Detail
Brand guidelines	Pending client provision (if available)
Colour palette	Client-provided or agency-proposed (to be confirmed)
Typography	System font stack for performance; client preferences honoured where possible
Imagery style	Minimal — the tool is utilitarian and data-driven, not marketing-led
Reference examples	None specified — to be discussed in Week 0 design kickoff
Elements to avoid	To be confirmed with client

SECTION 11

Hosting & Infrastructure

Per SOW responses 5.3 and 5.4, hosting approach and data residency are pending. This section captures the Fonix recommendation as a starting point; a final decision will be taken before Week 0 completes.

Requirement	Detail
Hosting provider (recommended)	AWS, Sydney region, to satisfy Australian data-residency preferences
Alternative options	DigitalOcean Sydney region; client-provided AWS/DO account if preferred
Domain	Client provides or Fonix can assist in procurement
SSL certificate	Included — Let's Encrypt or AWS Certificate Manager
CDN	CloudFront for static assets; not strictly required for MVP
Environments	Staging (Fonix-provided during build) and Production (client environment at handover, or Fonix continues hosting with cost pass-through)
Deployment ownership	Fonix deploys during build and handover; post-handover either Fonix-managed or client-managed based on preference
Data residency	Recommended: all data stored in AWS Sydney region (ap-southeast-2). Pending client confirmation
Estimated monthly hosting cost	$40-80 USD/month at launch volume, passed through transparently if Fonix hosts
Claude API cost (separate)	$200-600 USD/month at 200+ submissions, disclosed separately from hosting

SECTION 12

Compliance & Regulatory

This is an internal business tool for an Australian commercial entity. No specific regulatory compliance frameworks (HIPAA, PCI DSS, SOC 2) apply. The following considerations still apply as standard practice.

Item	Detail
Australian Privacy Act	Applies — the tool handles personal information of assessors and potentially property addresses. Standard privacy protections built in
Data residency	Australian data residency requested (per SOW 5.4 — pending confirmation). Recommended AWS Sydney region
Audit logging	All logins, uploads, downloads, and admin actions logged; retention aligned with job-data retention
Legal documentation	Client provides terms of service and privacy policy applicable to their business; Fonix does not provide legal documents
Data subject rights	Not applicable — the tool handles business data, not consumer PII

SECTION 13

Risk Register

ID	Risk	Likelihood	Impact	Mitigation	Owner
R-1	Client delays in providing additional sample submissions	Medium	High	Deadline set at kickoff; BA follows up proactively; build progresses on samples in hand	BA
R-2	New submission format encountered that our canonical schema doesn't handle (e.g., architect or NatHERS software we haven't seen)	High	Medium	Canonical schema built for extensibility; 20% timeline buffer reserved specifically for this	CTO
R-3	Claude API pricing or rate-limit changes during build	Low	Medium	Current rates locked at development start; alternative vision APIs evaluated as contingency	CTO
R-4	Plan-as-primary reconciliation direction proves suboptimal in edge cases	Medium	Medium	Reconciliation direction made configurable in code; can be tuned post-launch	CTO
R-5	Scope creep through additional requirements surfaced during UAT	Medium	Medium	Change request process (Section 14) enforces commercial boundary; UAT is for requirements validation, not new feature requests	BA
R-6	Hosting decision delayed beyond Week 0	Low	Low	Fonix-managed hosting is the fallback; work can proceed on staging while decision is pending	BA
R-7	Australian data residency requirement emerges late and forces region change	Low	Medium	Default recommendation is AWS Sydney region,	CTO

ID	Risk	Likelihood	Impact	Mitigation	Owner
				eliminating risk if client confirms	
R-8	Revision-comparison logic proves more complex than scoped (e.g., window renumbering across revisions)	Medium	Medium	Matching algorithm built with multiple strategies; 1 week allocated specifically for this module	Team Lead
R-9	Individual accounts and admin dashboard add complexity vs original MVP scope	Low	Low	Both are well-understood features; no novel technical risk	Team Lead
R-10	Client's 200+/month volume estimate proves low — actual volume is higher	Low	Medium	Horizontal scaling architecture; monitoring in place to detect capacity issues early	CTO

SECTION 14

Change Request Process

During the course of this project, new ideas and additional requirements may arise. This is natural and expected. To ensure these changes are handled transparently and do not disrupt the project timeline or budget, the following process applies:

1. **Any request outside the scope defined in this document** will be documented as a change request by the BA.
2. The project team will **assess the impact** of the change — additional effort, timeline impact, technical implications.
3. A written assessment will be shared, and **commercial terms discussed** if additional cost is involved.
4. **No out-of-scope work begins** until the change request is formally approved in writing.
5. Approved changes are added to the project scope with revised timelines and documentation updates.

This process protects both parties — ensuring the client always knows what they are getting and the delivery team always knows what they are building.

SECTION 15

Glossary

Terms used in this document that a non-technical stakeholder may find unfamiliar:

Term	Definition
API	Application Programming Interface — a way for two software systems to communicate
BASIX	Building Sustainability Index — a NSW government certificate showing a new home meets sustainability targets
BRD	Business Requirements Document — this document
BERS Pro	One of three government-accredited NatHERS assessment software tools
Claude	An AI assistant made by Anthropic; used here for document understanding and data extraction
Canonical schema	A single, consistent format into which data from many different source formats is translated
CRM	Customer Relationship Management system — tools used to track customer information and interactions
Confidence score	A number 0-100 representing how certain the system is that an extraction is correct
Consistency report	A summary sheet listing any disagreements between the plan and the certificates
FirstRate5	One of three government-accredited NatHERS assessment software tools
FPRD	Functional Product Requirements Document — a technical specification produced after this BRD
Hero	One of three government-accredited NatHERS assessment software tools (Chenath engine)
MoSCoW	A prioritisation method: Must have, Should have, Could have, Won't have
MVP	Minimum Viable Product — the first working version delivered to users
NatHERS	Nationwide House Energy Rating Scheme — a government certificate rating a home's thermal efficiency

Term	Definition
NFR	Non-Functional Requirement — a requirement about how the system performs, not what it does (e.g., speed, security)
RBAC	Role-Based Access Control — permissions granted based on the user's role
RPO	Recovery Point Objective — maximum acceptable data loss in the event of a disaster, expressed in time
RTO	Recovery Time Objective — maximum acceptable downtime in the event of a disaster
SHGC	Solar Heat Gain Coefficient — a measure of how much solar heat passes through a window
SOW	Statement of Work — the commercial contract document
Takeoff	The process of extracting quantities from plans (the core workflow being automated)
TPA	A specific commercial energy report format occasionally seen in Nathan's pipeline
U-value	A measure of how well a window transfers heat; lower values mean better insulation
UAT	User Acceptance Testing — the phase where the client tests the system before accepting it
WBS	Work Breakdown Structure — the detailed task list behind a project
XLSX	The file format for Microsoft Excel 2007+ workbooks

Thank you

Fonix Technologies looks forward to delivering this project. Any questions about this document can be directed to Lavesh.

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